



GATE ME 90 Days Challenge

Day 3/90 · Practice Set with Detailed Solutions

SUBJECT

**Engineering
Mathematics**

TODAY'S TOPIC

Linear Algebra

COUNTDOWN

108 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on matrix algebra, rank, systems of linear equations, eigenvalues and eigenvectors. Try them in **40 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

The rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$ is

(A) 0

(B) 1

(C) 2

(D) 3

Q2 (NAT)

The determinant of $\begin{bmatrix} 2 & 1 & 3 \\ 0 & 4 & 5 \\ 1 & 0 & 6 \end{bmatrix}$ is _____

Q3 (MCQ)

The system $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ has **no solution** when

(A) $\lambda = 3$, $\mu = 10$

(B) $\lambda \neq 3$, any μ

(C) $\lambda = 3$, $\mu \neq 10$

(D) $\lambda = 10$, $\mu = 3$

Q4 (NAT)

For $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$, the trace of A^3 is _____

Q5 (MCQ)

An eigenvector of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ corresponding to the eigenvalue 3 is

(A) $[1, -1]^T$

(B) $[1, 1]^T$

(C) $[1, 2]^T$

(D) $[2, 1]^T$

Q6 (NAT)

For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, A^4 can be written as $pA + qI$. The value of p is _____

Q7 (MCQ)

Which statement about a real symmetric matrix is **NOT** always true?

- (A) All its eigenvalues are real
(B) Eigenvectors for distinct eigenvalues are orthogonal
(C) It is invertible
(D) It can be diagonalised by an orthogonal matrix

Q8 (NAT)

A is a 4×5 real matrix of rank 3. The number of linearly independent solutions of $Ax = 0$ is _____

Q9 (MCQ)

If $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, the eigenvalues of $A^2 + 2A + I$ are

- (A) 3 and -1
(B) 9 and 1
(C) 16 and 0
(D) 4 and 0

Q10 (NAT)

A is a 3×3 matrix A has eigenvalues 1, 2 and 3. The determinant of $\text{adj}(A)$ is _____

Q11 (MCQ)

Solving $2x + y - z = 1$, $x + 3y + 2z = 13$, $3x - y + 4z = 13$ by Gauss elimination, the value of z is

- (A) 1
(B) 2
(C) 3
(D) 4

Q12 (NAT)

The rank of $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \\ 3 & 5 & 7 & 9 \\ 1 & 1 & 1 & 1 \end{bmatrix}$ is _____

Q13 (MCQ)

The system $kx + y = 0$, $x + ky = 0$ has a non-trivial solution when

- (A) $k = 0$ only
(B) $k = \pm 1$
(C) $k = \pm 2$
(D) no real k

Q14 (NAT)

The largest eigenvalue of $\begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$ is _____

Q15 MCQ

The matrix $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ has

- (A) two linearly independent eigenvectors (B) only one linearly independent eigenvector
(C) no real eigenvalue (D) eigenvalues 1 and 0
-

Answer Key

Q	Type	Answer
Q1	MCQ	(C) 2
Q2	NAT	41
Q3	MCQ	(C) $\lambda = 3, \mu \neq 10$
Q4	NAT	251
Q5	MCQ	(B) $[1, 1]^T$
Q6	NAT	145
Q7	MCQ	(C) It is invertible
Q8	NAT	2
Q9	MCQ	(C) 16 and 0
Q10	NAT	36
Q11	MCQ	(C) 3
Q12	NAT	2
Q13	MCQ	(B) $k = \pm 1$
Q14	NAT	5
Q15	MCQ	(B) only one linearly independent eigenvector

Detailed Solutions

Q1 Answer: (C) 2

Concept: Rank = number of non-zero rows after row reduction = order of the largest non-zero minor.

$R_2 = 2R_1$, so $R_2 \rightarrow R_2 - 2R_1$ gives a zero row. $\det A = 0$, so rank < 3 .

R_1 and R_3 are not proportional: the minor $\begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1 - 2 = -1 \neq 0$.

Rank = 2

Exam tip: Look for proportional rows or columns first; it often settles the rank without full elimination.

Q2 Answer: 41

Concept: Expand along a row or column with zeros to save work.

Along row 1: $2(4 \cdot 6 - 5 \cdot 0) - 1(0 \cdot 6 - 5 \cdot 1) + 3(0 \cdot 0 - 4 \cdot 1)$

$= 2(24) - 1(-5) + 3(-4) = 48 + 5 - 12 = 41$

Exam tip: Watch the sign pattern $+ - +$ along the first row; the middle term flips sign.

Q3 Answer: (C) $\lambda = 3, \mu \neq 10$

Concept: Compare $\text{rank}(A)$ with rank of the augmented matrix $[A | b]$. Unequal ranks $\rightarrow$ no solution.

$R_3 - R_2$ gives $(0, 0, \lambda - 3 | \mu - 10)$.

$\lambda \neq 3$: $\text{rank } A = 3 = \text{rank } [A|b] \rightarrow$ unique solution.

$\lambda = 3, \mu = 10$: both ranks 2 $\rightarrow$ infinitely many solutions.

$\lambda = 3, \mu \neq 10$: $\text{rank } A = 2, \text{rank } [A|b] = 3 \rightarrow$ **no solution**.

Exam tip: Reduce only the row that contains λ and μ ; the other rows need no work.

Q4 Answer: 251

Concept: If λ is an eigenvalue of A then λ^3 is an eigenvalue of A^3 , and $\text{trace} = \text{sum of eigenvalues}$.

A is lower triangular, so its eigenvalues are the diagonal entries 2, 3, 6.

Eigenvalues of A^3 : 8, 27, 216

$\text{trace}(A^3) = 8 + 27 + 216 = \mathbf{251}$

Exam tip: Never multiply out A^3 in the exam. Triangular $\rightarrow$ read eigenvalues from the diagonal.

Q5 Answer: (B) $[1, 1]^T$

Concept: Solve $(A - \lambda I)x = 0$ for $\lambda = 3$.

$A - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \rightarrow -x_1 + x_2 = 0 \rightarrow x_1 = x_2$

Eigenvector: $[1, 1]^T$ (any non-zero multiple).

Check: $A[1, 1]^T = [3, 3]^T = 3[1, 1]^T \checkmark$. $[1, -1]^T$ belongs to $\lambda = 1$.

Exam tip: In an MCQ, just multiply each option by A and see which comes out as λ times itself.

Q6 Answer: 145

Concept: Cayley-Hamilton: every square matrix satisfies its own characteristic equation.

Characteristic equation: $\lambda^2 - (\text{trace})\lambda + \det = \lambda^2 - 5\lambda - 2 = 0 \rightarrow A^2 = 5A + 2I$

$A^3 = 5A^2 + 2A = 5(5A + 2I) + 2A = 27A + 10I$

$A^4 = 27A^2 + 10A = 27(5A + 2I) + 10A = 145A + 54I$

$p = \mathbf{145}$ (and $q = 54$)

Exam tip: Cayley-Hamilton also gives the inverse: $A^{-1} = (A - 5I)/2$ here.

Q7 Answer: (C) It is invertible

Concept: The spectral theorem guarantees real eigenvalues, orthogonal eigenvectors and orthogonal diagonalisation for every real symmetric matrix.

Invertibility needs every eigenvalue to be non-zero, which is not guaranteed.

Counter-example:

$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ is symmetric but $\det = 0$ (eigenvalues 0 and 2).

So '**It is invertible**' is the false statement.

Exam tip: Symmetric says nothing about singularity; a zero eigenvalue is allowed.

Q8 Answer: 2**Concept:** Rank-nullity theorem: rank + nullity = number of unknowns (columns).Unknowns $n = 5$, rank $r = 3$ Nullity = $n - r = 5 - 3 = 2$ **Exam tip:** Use the number of columns, not rows. $4 - 3 = 1$ is the usual slip.**Q9 Answer: (C) 16 and 0****Concept:** If $Ax = \lambda x$ then $f(A)x = f(\lambda)x$ for any polynomial f .Eigenvalues of A : $\lambda^2 - 2\lambda - 3 = 0 \rightarrow \lambda = 3, -1$ $f(\lambda) = \lambda^2 + 2\lambda + 1 = (\lambda + 1)^2 \rightarrow (3 + 1)^2 = 16$ and $(-1 + 1)^2 = 0$ Eigenvalues of $A^2 + 2A + I$: **16 and 0****Exam tip:** Note that $A^2 + 2A + I$ is singular here, because one of its eigenvalues is zero.**Q10 Answer: 36****Concept:** $\det(\text{adj } A) = (\det A)^{n-1}$ for an $n \times n$ matrix. $\det A = \text{product of eigenvalues} = 1 \times 2 \times 3 = 6$ $\det(\text{adj } A) = 6^{3-1} = 6^2 = 36$ **Exam tip:** Also remember $\text{adj}(\text{adj } A) = (\det A)^{n-2} A$.**Q11 Answer: (C) 3****Concept:** Eliminate x , then y , then back-substitute. $R_2 \rightarrow 2R_2 - R_1: 5y + 5z = 25 \rightarrow y + z = 5$ $R_3 \rightarrow 2R_3 - 3R_1: -5y + 11z = 23$ Adding $5(y + z = 5): 16z = 48 \rightarrow z = 3$, then $y = 2$ and $x = 1$.**Exam tip:** Substitute (1, 2, 3) back into all three equations. It takes 10 seconds and catches sign errors.**Q12 Answer: 2****Concept:** Rank = number of linearly independent rows. $R_2 = 2R_1$, so R_2 adds nothing. $R_3 = 2R_1 + R_4: 2[1, 2, 3, 4] + [1, 1, 1, 1] = [3, 5, 7, 9] \checkmark$, so R_3 adds nothing either. R_1 and R_4 are not proportional $\rightarrow \text{rank} = 2$ **Exam tip:** Rows whose entries are in arithmetic progression (1 2 3 4, 3 5 7 9, 1 1 1 1) always give rank ≤ 2 .**Q13 Answer: (B) $k = \pm 1$** **Concept:** A homogeneous system has non-trivial solutions only if the coefficient matrix is singular. $\det = k^2 - 1 = 0 \rightarrow k = \pm 1$ Check $k = 1: x + y = 0$ twice $\rightarrow x = -y$, infinitely many solutions $\checkmark$ **Exam tip:** Homogeneous systems are always consistent; the only question is whether $\det = 0$.

Q14 Answer: 5

Concept: $A = 2I + J$, where J is the all-ones matrix. J has eigenvalues 3, 0, 0.

So A has eigenvalues $2 + 3 = 5$, $2 + 0 = 2$, $2 + 0 = 2$.

Largest = **5**. Check: trace = $9 = 5 + 2 + 2$ ✓; det = $20 = 5 \times 2 \times 2$ ✓

Exam tip: Equal row sums s mean s is an eigenvalue with eigenvector $[1, 1, 1]^T$. Here every row sums to 5.

Q15 Answer: (B) only one linearly independent eigenvector

Concept: Number of independent eigenvectors for $\lambda = n - \text{rank}(A - \lambda I)$ (geometric multiplicity).

Eigenvalue $\lambda = 1$ is repeated (algebraic multiplicity 2).

$A - I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has rank 1 $\rightarrow$ independent eigenvectors = $2 - 1 = 1$ (only $[1, 0]^T$).

So A is not diagonalisable.

Exam tip: A repeated eigenvalue does not guarantee a full set of eigenvectors; check the rank of $A - \lambda I$.

What next?

Day 3 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



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